MVP

LURNEX

AI-Guided Mastery System

Product Masterplan · v1.0

CONFIDENTIAL · SOLO PROJECT · 2025

01 Executive Summary

Lurnex is an AI-guided mastery system — not an AI tutor, not a chatbot with folders. It is a structured learning operating system that tracks what you actually understand, not just what you clicked or watched.

WHAT EXISTING TOOLS DO	WHAT LURNEX DOES
Track content completion	**Infer cognitive state**
Show quiz scores	**Map understanding depth across 5 levels**
One chat for everything	**Isolated spaces: learn, doubt, test**
Static learning paths	**Adaptive flowchart that unlocks and blocks dynamically**
Soft, encouraging AI	**Strict AI that catches false confidence**

9/10	7/10	8/10	8/10
Problem clarity	Differentiation	Buildability	Market timing

02 The Problem

Knowledge is abundant. Navigating it effectively is not.

— **One chat becomes a mess**
Uploads, doubts, learning, and rabbit holes all collide in the same thread. Finding anything later is impossible.

— **No structure, no progression**
AI tools answer questions but never enforce a learning path. You can jump anywhere, learn nothing deeply.

— **False confidence goes undetected**
Students say 'I understood' and move on. No tool catches whether they actually did.

— **Scores, not understanding**
Quizzes test recognition, not transfer. You can ace a quiz and still not understand the mechanism.

— **Off-topic drift**
Ask an AI mid-course about something unrelated and it answers. No enforcement. No focus.

03 Cognitive Model

The system tracks understanding at 5 levels, not just completion.

LEVEL	NAME	WHAT IT MEANS	STATE

L1	**Recognition**	Seen the concept before	SURFACE
L2	**Mechanical**	Can solve standard problems	PARTIAL
L3	**Conceptual**	Understands why it works	FUNCTIONAL
L4	**Transfer**	Can apply it in new contexts	STRONG
L5	**Intuitive**	Predicts behavior without solving	MASTERED

Cognitive signals the AI tracks

Repeated questioning Asking the same concept 3 different ways = conceptual instability detected	**False confidence** User says 'I got it' then fails transfer questions = surface grasp flagged
Hint dependency High hint usage on standard problems = mechanical level, not conceptual	**Concept switching** Rapid topic jumps without resolution = avoidance pattern, not mastery
Mistake similarity Repeated same error type across sessions = deep misconception identified	**Recall delay** Long pauses before correct answer = fragile, not fluent understanding

Topic color states (replaces numerical scores)

UNSTABLE Foundation broken or missing. Prerequisites need revisiting.	PARTIAL Mechanical understanding only. No depth or transfer.	FUNCTIONAL Conceptual grasp. Understands why, not just how.	MASTERED Intuitive. Can transfer to new problems spontaneously.

04 Platform Structure

Entry: two modes

MODE A — SOURCE GROUNDED	MODE B — AI AS TEACHER
Upload PDFs, slides, links, or notes. Describe your goal and pace. AI builds the learning plan strictly from your uploaded material, cites sources throughout, and stays inside them. More accurate, less hallucination.	**Describe the topic and your goals. No uploads needed.** AI generates curriculum from its own knowledge and teaches fully. Trade-off: hallucination risk. This is surfaced clearly to the user at setup. HALLUCINATION RISK

Three-panel main interface

After setup, the entire learning experience lives in a single three-panel layout. Each panel has a strict, non-overlapping role.

LEFT PANEL	MIDDLE PANEL	RIGHT PANEL
Mini roadmap Shows the topic tree for the current section. Topics are color-stated (done / active / locked / unstable). Clicking navigates. Locked topics show the prerequisite you're missing. Persistent nudge at bottom showing weakest connection in your knowledge graph.	**Structured lesson — paginated** AI-authored content only. No chat input. Navigate with prev / next like a book. Page-level controls: Simplify, Go Deeper, Add Example, Export. Select any text to trigger inline rewrite bar. AI rewrites just that fragment in-place. Undo available for 10 seconds.	**Doubt chat — scoped** Context-locked to the current page. The only place you type freely. Off-topic questions get a redirect, not an answer. If you ask about a topic three chapters ahead: 'That comes when you reach backprop. Stay here first.'

Inline text rewrite

1	Select any text on the lesson page
2	Floating mini bar appears with presets: Explain deeper / Add example / Simplify / Add analogy — plus free text input
3	AI rewrites only the selected fragment in-place. Full page sent as context so rewrite fits naturally with surrounding content.
4	Undo button appears for 10 seconds to restore original fragment. Rewrite becomes part of the page for export.

Export options

Full course PDF	Topic notes	Markdown	Flashcards	Notion (v2)

05 Adaptive Flowchart

The roadmap is not static. It unlocks, blocks, and restructures based on your understanding.

UNLOCK	Master vector spaces + transformations → PCA auto-unlocked, linear algebra sections accelerated
SLOW DOWN	Struggle with recursion + stack memory → Trees delayed, DP postponed, recursion reinforcement inserted
INTERVENE	False confidence detected mid-topic → AI blocks progression, surfaces the exact gap, requires re-test before advancing
BRIDGE	Isolated node detected (topic learned but not connected) → Bridge lesson generated teaching it in relation to connected topics

Progression is never binary. Understanding is layered — the flowchart reflects actual depth, not just topic completion. A topic can be marked 'done' and later downgraded to 'unstable' if subsequent sessions reveal shallow understanding.

06 Knowledge Connection Graph

Shows how well you have connected topics — not just whether you learned them individually.

The connection graph is a separate dedicated view, not embedded in the main layout. It has two layers rendered simultaneously:

Structural layer (fixed)	**Mastery layer (live)**
Built at course creation. Does not change.	Updates after quizzes, doubt sessions, and topic completion.
Nodes = topics. Edges = conceptual dependencies between them. Clusters = topic groups. This is the ground truth of how the subject is connected.	Node color = your understanding state. Edge thickness = how well you connected two topics together, not just individually. Isolated nodes = learned in isolation, never applied in relation to adjacent concepts.

The isolated node insight

A node can be green (mastered) but its edge to another node can be thin — meaning you learned both topics but never connected them. You understood linear regression. You understood gradient descent. But you've never applied gradient descent to linear regression in a problem — so the edge between them is weak. This is surface learning made visible.

Edge strength is built from three signals:

— Transfer questions answered correctly that crossed both topics simultaneously
— Doubt chat questions that showed you linking the two concepts
— Quiz problems that tested them together, not in isolation

Clicking any node shows:

Your understanding state for that topic Which edges are strong vs weak Why a connection is missing	**Action buttons:** Reconnect this topic — starts a bridge lesson Quiz on connections — tests cross-topic transfer See where it connects — highlights all edges

07 Quiz System

The most important feature. The only source of ground truth about what you actually understand.

Everything else in Lurnex is inference. The lesson is delivery. The doubt chat is support. The quiz is measurement. It gets more engineering investment than anything else.

Three question types (no multiple choice)

TYPE	FORMAT	LEVEL TESTED
Apply	Given a new scenario, apply the concept. 'Your model has high training accuracy but low test accuracy. What does this tell you about regularization?'	TRANSFER L4
Spot the error	Here is a flawed reasoning. Identify what's wrong and why. Forces understanding of the mechanism, not just the procedure.	CONCEPTUAL L3
Explain it	Explain X to someone who hasn't seen it. Tests whether you can reconstruct the idea, not just recognize it.	INTUITIVE L5

How quiz results update the roadmap

PASS WITH TRANSFER	Topic moves to Mastered. Adjacent locked topics unlocked if prerequisites met.
PASS MECHANICAL ONLY	Topic stays Partial. Conceptual and transfer questions queued for next session.
FAIL AFTER CLAIMING UNDERSTOOD	False confidence flagged. Topic downgraded. AI surfaces the specific gap and blocks progression.
FAIL PREREQUISITE PATTERN	System traces root cause. If it's a prerequisite gap, that earlier topic is marked Unstable and revisit is scheduled.

08 Token Architecture & Cost Engineering

Core principle: generate once, store everything, infer lazily.

The naive implementation (treating every background process as a live completion) would make per-user costs unsustainable. The architecture below keeps costs low without sacrificing capability.

Model tier strategy

TASK	MODEL	WHEN	WHY
Course plan, lesson generation, graph structure	**Opus**	Once at setup	Quality matters. Not repeated.
Doubt chat, inline rewrite, quiz questions	**Sonnet**	Per interaction	Good enough, 5x cheaper than Opus.
Mastery inference, relevance check, false confidence detection	**Haiku**	Background, per message	Classification tasks. Haiku handles perfectly.

Engineering optimizations

Batch generation	Generate all lesson pages upfront at course setup using the Anthropic Batch API (50% cheaper). User isn't waiting live.
Pre-generated quiz pool	Generate 20-30 questions per topic at setup. Serve from pool. Regenerate only when pool is low or mastery state shifts significantly.
Paragraph-level chunking	For inline rewrite, don't send the full page — send the selected fragment plus 2-3 surrounding paragraphs. Enough context, fraction of tokens.
Piggyback inference	Mastery state update runs on the same call as the doubt chat response using structured output. One call, two jobs.
Trigger-based graph update	Connection graph edges update on triggers only: quiz completed, topic done, or doubt session ends. Not on every message.
Lazy cognitive tracking	Full cognitive signal analysis runs once per session end, not per message. Batch the signals, infer once.

~30 doubt messages per active hour	**~5 inline rewrites** per active hour	**1 quiz session** per topic	**$0.05 – $0.15** per active user hour (tiered model)

09 MVP Scope

Ship the core loop. Validate before building the intelligence layer.

The MVP is three things done well: structured AI lesson + scoped doubt chat + honest quiz. Everything else is post-MVP. Speed of shipping matters more than feature completeness right now.

What's in the MVP

✓ Mode B (AI as teacher) — easier to build, no file processing required. Mode A (source upload) in v2.
✓ AI lesson generation — paginated middle panel, page-level controls (Simplify / Go Deeper)
✓ Scoped doubt chat — right panel, context-locked to current page
✓ Basic quiz — 3 question types (apply, spot error, explain). 10 questions generated per topic at setup.
✓ Roadmap state update after quiz — binary done/not-done for MVP, full color states in v2
✓ Mini roadmap in left panel — topic tree navigation, locked/unlocked states

What's post-MVP

— Connection graph and edge strength tracking
— Inline text rewrite (select + mini prompt bar)
— False confidence detection and misconception interceptor
— Full 5-level cognitive tracking and color states
— Mode A — source upload and citation
— Export to PDF / Markdown / Flashcards
— Spaced repetition and recall scheduling
— Bridge lessons for isolated nodes

Build order

DAY	TASK
1	Setup + onboarding flow. Topic input, goal description, AI generates lesson plan and page content in batch.
2	Three-panel layout. Left roadmap, middle paginated lesson, right doubt chat. Navigation working.
3	Quiz system. Question generation at setup, serving from pool, basic roadmap state update on completion.
4	Polish + context locking. Doubt chat scoped to page. Relevance redirect. Page-level controls.
5	Testing, edge cases, and basic styling. Demo-ready.

10 Positioning & Moat

Not 'AI tutor'. Not 'AI learning app'.

AI-Guided Mastery System	This framing justifies premium positioning, enterprise/academic licensing, and separates Lurnex from the crowded 'chat with your notes' category. The product is about structure, progression, dependency enforcement, and cognitive integrity — not chatting with an AI about slides. The expensive AI tutoring tools price themselves out of students who need them most. Being the affordable serious option is both a market position and the right thing to do.

Why this has a moat

Relevance enforcement	Technically and UX-wise hard to get right. Most competitors skip it entirely.
Connection graph	A genuinely novel signal. No major platform tracks conceptual edge strength between topics.
Isolated context spaces	Solves a problem users feel but can't articulate. High retention insight.
Source-grounded teaching	Students trust it more than a generic LLM tutor. Citations build credibility.
Long-term knowledge state	Once a user builds their graph, mastery map, and learning history, switching is painful.

Competitive landscape

TOOL	WHAT IT DOES	WHAT'S MISSING
Khanmigo	AI tutor for Khan content	No mastery tracking, no structure, expensive
ChatGPT	Answers questions	No progression, no enforcement, no memory
Notion AI	Notes + AI search	Not a learning system at all
Anki	Spaced repetition flashcards	No understanding model, just recall
Lurnex	Full mastery OS	— (this is the gap)

11 Open Questions for Next Planning Session

1	**Quiz placement** Does the quiz appear as a separate tab, or inline after each topic completes in the main view?
2	**Two-level navigation** How does the top-level roadmap (full course overview) connect down to the mini roadmap in the left panel? Needs to feel seamless, not like two separate tools.
3	**Unstable trigger** What exactly triggers a topic being marked Unstable — quiz failure, AI detection from doubt chat patterns, time decay, or a combination of all three?
4	**Pricing tiers** Standard plan: one quiz generation per topic. Higher plans: more flexibility. But keep it genuinely affordable for students — monetize institutions, not individuals.
5	**Mode A scope** File processing complexity (PDFs, slides, links). What's the minimum viable source parser for v1.1?
6	**Solo or team** MVP is buildable solo in a week. The question is what happens at week 3 when motivation dips. Accountability structure needed, not necessarily a co-founder.

This document is a living plan. Every section should be treated as a draft until the MVP ships and real user behavior tells you what to keep, cut, and change.

Build it. Ship it. Learn from it.